

Electric Charge

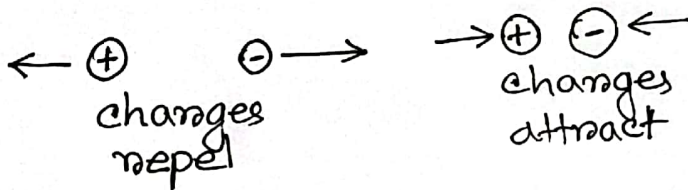
- * What is Electric Charge
- * How do electric charges interact?

Excess charge: The net charge is also sometimes called excess charge because a charged object has an excess of either positive or negative charges.

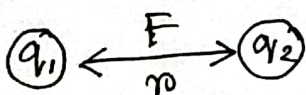
Static Electricity:

A tiny imbalance in either positive or negative charge on an object is the case of static electricity.

	Electron	Proton	Neutron
Mass (kg)	9.109×10^{-31}	1.673×10^{-27}	1.675×10^{-27}
charge (coulombs)	-1.602×10^{-19}	$+1.602 \times 10^{-19}$	0



Coulomb's Law



$$\vec{F} = k \frac{q_1 q_2}{r^2} \hat{r}$$

q_1, q_2 = the amount of two charges

F = Electrostatic force between two

r = distance between two charges.

$k = \frac{1}{4\pi\epsilon_0}$ = Coulombs Constant

$$k = 8.987752 \times 10^9 \text{ Nm}^2/\text{C}^2$$

$$k = 8.99 \times 10^9 \text{ Nm}^2/\text{C}^2 \text{ or } 9 \times 10^9 \text{ Nm}^2/\text{C}^2$$

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{N}\cdot\text{m}^2$$

Coulomb's Law;

- * The force between two charges gets stronger as the charges move closer together.
- * The force also gets stronger if the amount of charge becomes larger.
- * The force between two charges is directed along the line connecting their centers.
- * Electric forces always occur in pairs according to Newton's third law, like all the forces.
- * The force between charges is directly proportional to the magnitude or amount of each charge.
- * Doubling one charge doubles the force.
- * Doubling both charges quadruples the force.

* The force between charges is inversely proportional to the square of the distance between them.

* Doubling the distance reduces the force by a factor of $(2)^2 = 4$, decreasing the force to one-fourth its original value ($1/4$).

* This relationship is called an **inverse square Law**, because force and distance follow an inverse square relationship.

Problem-1

A positive charge $6 \times 10^{-6} \text{ C}$ is 0.040 m from the second positive charge of $4 \times 10^{-6} \text{ C}$. Calculate the force...

$$\vec{F} = k \frac{q_1 q_2}{r^2} \hat{n} \quad q_1 = +6 \times 10^{-6} \text{ C}$$

$$= 9 \times 10^9 \frac{(6 \times 10^{-6} \times 4 \times 10^{-6})}{(0.04)^2} \hat{n} \quad q_2 = +4 \times 10^{-6} \text{ C}$$

$$= 135 \text{ N}$$

Problem: 2

If the distance between the two charges are doubled what is the change in the force between them?

Problem: 3

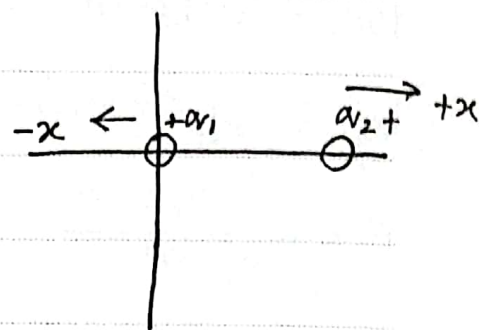
Two positively charged particles fixed in place on an x-axis. The $q_1 = 1.60 \times 10^{-19} \text{ C}$ and $q_2 = 3.20 \times 10^{-19} \text{ C}$, particle separation is $R = 0.02 \text{ m}$. What is the magnitude and direction of the electrostatic force $\vec{F}_{12}$ on particle 1 from particle 2?

The q_1 will experience force along $-x$ axis. $(-i)$.

$$\vec{F}_{12} = k \frac{q_1 q_2}{r_{12}^2} \hat{i}$$

$$= 9 \times 10^9 \frac{(1.60 \times 10^{-19} \times 3.20 \times 10^{-19})}{(0.02)^2} \hat{i}$$

$$= 1.152 \times 10^{-24} \hat{i}$$



Thus the q_1 will experience force along with $-x$ axis. So, we can write $\vec{F}_{12} = -(1.152 \times 10^{-24}) \hat{i}$

Problem: 4

Two charges $q_1 = 5 \times 10^{-12} \text{ C}$ and $q_2 = 3 \times 10^{-15} \text{ C}$ are separated at 3 mm distance. What will be the electrostatic force?

$$\vec{F} = k \frac{q_1 q_2}{r^2}$$

$$= 9 \times 10^9 \frac{(5 \times 10^{-12} \times 3 \times 10^{-15})}{(3 \times 10^{-3})^2}$$

$$= 1.5 \times 10^{-11} \text{ N}$$

$$q_1 = 5 \times 10^{-12} \text{ C}$$

$$q_2 = 3 \times 10^{-15} \text{ C}$$

$$r = 3 \text{ mm} = 3 \times 10^{-3} \text{ m}$$

Problem: 5

The nucleus of an iron atom has a radius of about $4.0 \times 10^{-15} \text{ m}$ and contains 26 protons.

① What is the magnitude of the repulsive electrostatic force between two of the protons that are separated by $4.0 \times 10^{-15} \text{ m}$?

$$F = k \frac{q_1 q_2 e^2}{r^2}$$

$$= 9 \times 10^9 \frac{(1.602 \times 10^{-19})^2}{(4.0 \times 10^{-15})^2}$$

$$= 14.4 \text{ N}$$

$$e = 1.602 \times 10^{-19} \text{ C}$$

$$r = 4.0 \times 10^{-15} \text{ m}$$

② What is the magnitude of the gravitational force between those same two protons?

$$F = G \frac{m_p^2}{r^2}$$

$$= 6.67 \times 10^{-11} \frac{1.67 \times 10^{-27}}{(4.0 \times 10^{-15})^2}$$

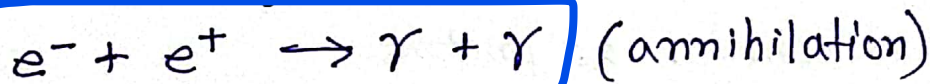
$$= 1.2 \times 10^{-35} \text{ N}$$

$$m_p = 1.67 \times 10^{-27} \text{ kg mass of proton}$$

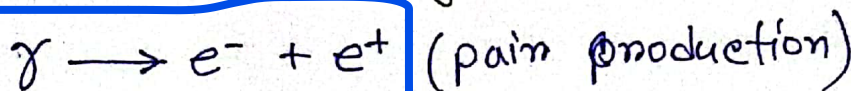
$$G = 6.67 \times 10^{-11} \text{ N m}^2 / \text{kg}^2$$

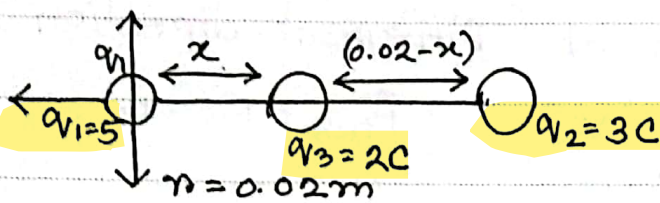
Charge Conservation:

Charge conservation occurs when an e^- ~~exchange~~ and its antiparticle, the positron e^+ ~~exchange~~, undergo an annihilation process, transforming into two gamma rays. (High energy light)



In pair production, the converse of annihilation, ~~charge~~ is also conserved. Gamma rays transform into an e^- , e^+





$$\vec{F} = 3.37 \times 10^{14} (-\hat{i}) \text{ for } q_1 \leftarrow$$

$$\vec{F} = 3.37 \times 10^{14} (\hat{i}) \text{ for } q_2 \rightarrow$$

Both charges are same, so they act as repulsive.

$$F = k \frac{q_1 q_2}{r^2}$$

$$= 9 \times 10^9 \frac{5 \times 3}{(0.02)^2}$$

$$= 3.375 \times 10^{14} \text{ N}$$

Vector Addition:

$$\vec{F}_1 + \vec{F}_2 = 0 = F_{\text{Net}}$$

$$\Rightarrow k \frac{10}{x^2} \hat{i} + k \frac{6}{(0.02-x)^2} (-\hat{i}) = 0$$

$$\vec{F}_1 = k \frac{q_1 q_3}{x^2} = k \frac{10}{x^2} \hat{i}$$

$$\vec{F}_2 = k \frac{q_1 q_2}{(0.02-x)^2} = k \frac{6}{(0.02-x)^2} (-\hat{i})$$

$$\Rightarrow \frac{10}{x^2} - \frac{6}{(0.02-x)^2} = 0$$

$$\Rightarrow \frac{10(0.02-x)^2 - 6x^2}{x^2(0.02-x)^2} = 0$$

$$\Rightarrow 10(0.02-x)^2 - 6x^2 = 0$$

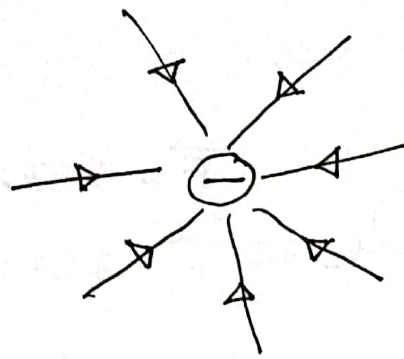
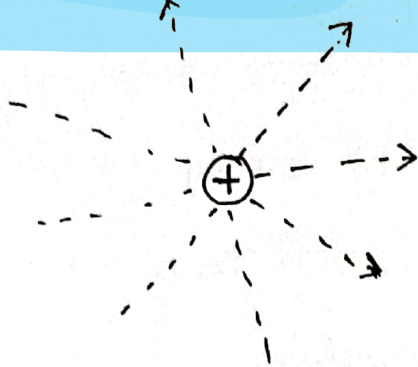
$$\Rightarrow 4 \times 10^{-3} + 0.4x - 10x^2 - 6x^2 = 0$$

$$\Rightarrow 4 \times 10^{-3} + 0.4x - 16x^2 = 0$$

$$\Rightarrow -16x^2 + 0.4x + 4 \times 10^{-3} = 0$$

$$\therefore x_1 = 0.033 \text{ m}, x_2 = -7.66 \times 10^{-3} \text{ m}$$

Electric Field

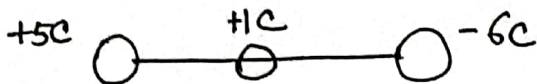


same direction $\Rightarrow$

$$F_{\text{Net}} = F_1 + F_2$$

Different direction $\Rightarrow$

$$F_{\text{Net}} = F_1 - F_2$$



$$F_{\text{Net}} = F_1 + F_2 = 0$$

$$\Rightarrow k_1 \frac{5}{x^2} + \frac{k_6}{(x-2)^2} = 0$$

$$\Rightarrow \frac{5}{x^2} + \frac{6}{(x-2)^2} = 0$$

$$\Rightarrow 5(x-2)^2 + 6x^2 = 0$$

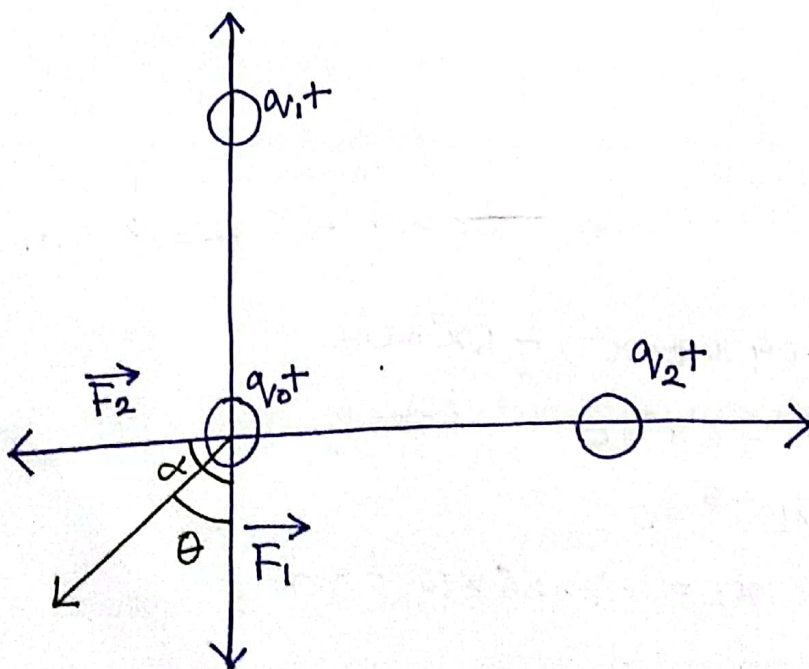
$$\Rightarrow 5(x^2 - 4x + 4) + 6x^2 = 0$$

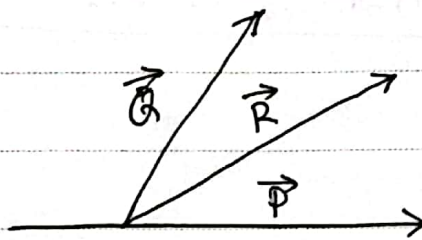
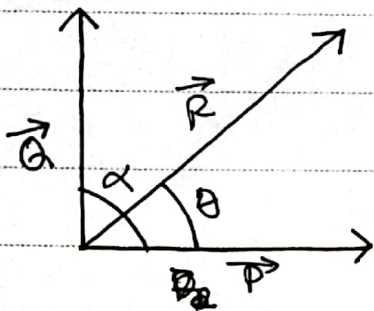
$$\Rightarrow 5x^2 - 20x + 20 + 6x^2 = 0$$

$$\Rightarrow 11x^2 - 20x + 20 = 0$$

$x \Rightarrow$

There is no place where.....





$$R = \sqrt{P^2 + Q^2 + 2PQ \cos \alpha}$$

$$\tan \theta = \frac{Q \sin \alpha}{P + Q \cos \alpha}$$

when P and Q are at 90°

$$R = \sqrt{P^2 + Q^2}$$

$$F_{\text{net}} = \sqrt{F_1^2 + F_2^2}$$

when P and Q $\alpha = 0^\circ$

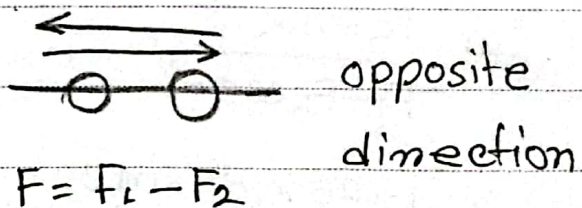
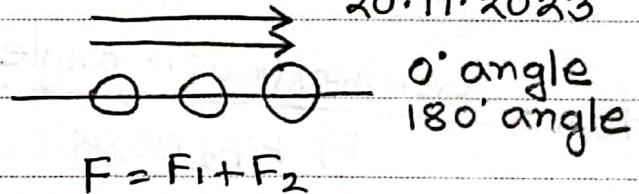
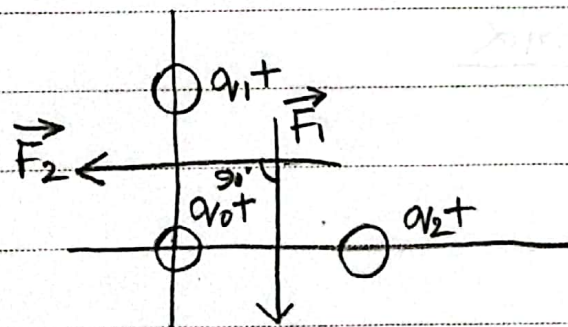
$$R = P + Q$$

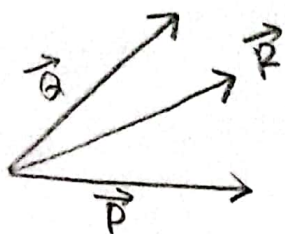
when P and Q $\alpha = 180^\circ$

$$R = P - Q$$

Physics, class: 15

20.11.2023





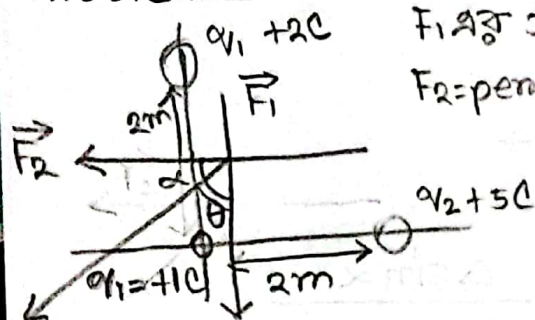
Resultant $R = \sqrt{p^2 + q^2 + 2pq \cos \alpha}$

$\alpha = 90^\circ$

$\cos \alpha = \cos 90^\circ = 0$

$F_{\text{Net}} = \sqrt{F_1^2 + F_2^2}$

Problem: 1



F_1 & F_2 on q_1 & q_2
 F_2 = perpendicular angle
 and vice versa

$$F_1 = k \frac{q_1 q_2}{r^2}$$

$$= 9 \times 10^9 \frac{2 \times 1}{(2)^2}$$

$$= 4.5 \times 10^9 \text{ N}$$

$$F_2 = k \frac{q_1 q_2}{r^2}$$

$$= 9 \times 10^9 \frac{1 \times 5}{(2)^2}$$

$$= 1.125 \times 10^{10} \text{ N}$$

$$\therefore F_{\text{Net}} = \sqrt{F_1^2 + F_2^2}$$

$$= \sqrt{(4.5 \times 10^9)^2 + (1.125 \times 10^{10})^2}$$

$$= 1.21 \times 10^{10}$$

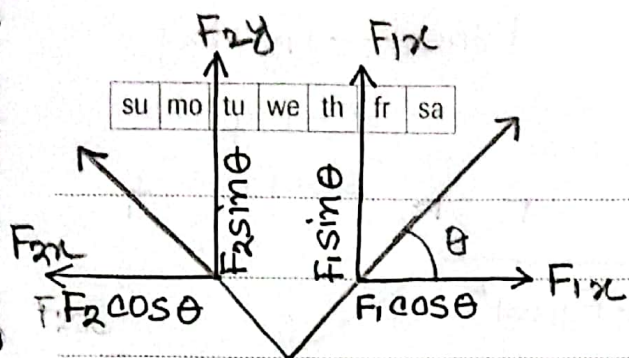
$$\theta = \tan^{-1} \frac{\text{perpendicular angle} \times \sin \alpha}{F_1 + F_2 \cos \alpha}$$

$$\therefore \theta = \tan^{-1} \frac{F_2 \sin \alpha}{F_1 + F_2 \cos \alpha}$$

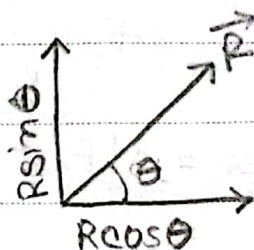
$$= \tan^{-1} \frac{1.125 \times 10^{10} \sin 90^\circ}{4.5 \times 10^9 + 1.125 \times 10^{10} \cos 90^\circ}$$

$$= \tan^{-1} \frac{1.125 \times 10^{10}}{4.5 \times 10^9}$$

$$= 68.199^\circ \quad (\text{Ans})$$



Date: / /

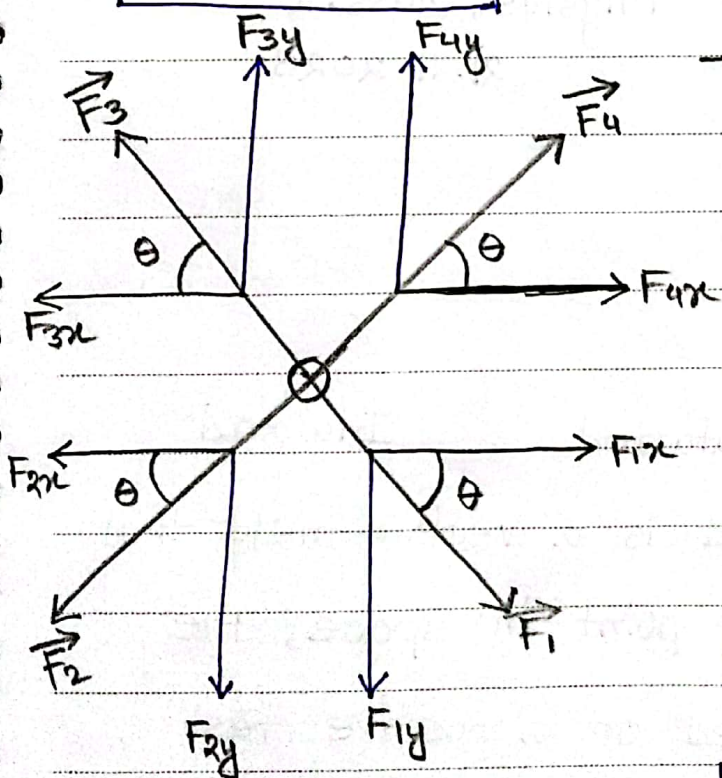
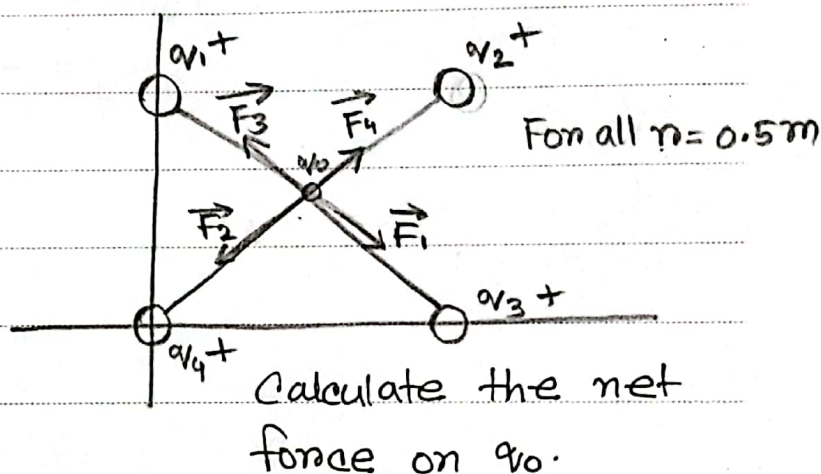


$$F_{x\text{net}} = F_{1x} - F_{2x} \rightarrow F_{x\text{net}} = F_1 \cos \theta - F_2 \cos \theta$$

$$F_{y\text{net}} = F_{1y} + F_{2y} \rightarrow F_{y\text{net}} = F_1 \sin \theta + F_2 \sin \theta$$

$$F_{\text{net}} = \sqrt{F_{x\text{net}}^2 + F_{y\text{net}}^2}$$

$$\tan \theta = \frac{F_{y\text{net}}}{F_{x\text{net}}}$$



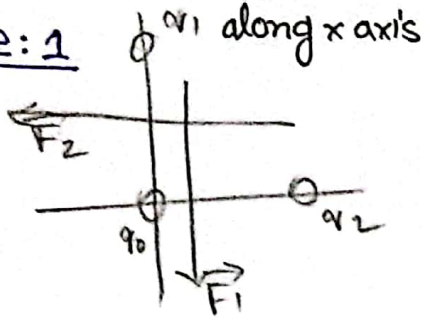
$$F_{x\text{net}} = F_{1x} + F_{4x} - F_{2x} - F_{3x}$$

$$F_{y\text{net}} = F_{3y} + F_{4y} - F_{1y} - F_{2y}$$

$$F_{\text{net}} = \sqrt{F_{x\text{net}}^2 + F_{y\text{net}}^2}$$

$$\tan \theta = \frac{F_{y\text{net}}}{F_{x\text{net}}} \quad \left[\begin{array}{l} \text{positive sign indicates} \\ \text{the degree angle} \end{array} \right]$$

There is no θ create
case: 1

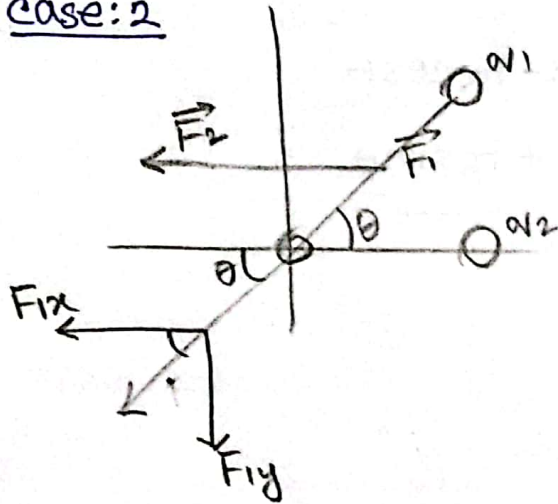


$$\begin{aligned} F_{x\text{Net}} &= F_{1x} - F_{2x} \\ &= -F_{2x} \\ &= -F_2 \end{aligned}$$

$$\begin{aligned} F_{y\text{Net}} &= -F_{1y} + F_{2y} \\ &= -F_{1y} \\ &= -F_1 \end{aligned}$$

$$F_{\text{Net}} = \sqrt{F_{x\text{Net}}^2 + F_{y\text{Net}}^2}$$

case: 2

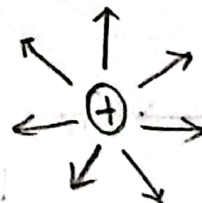
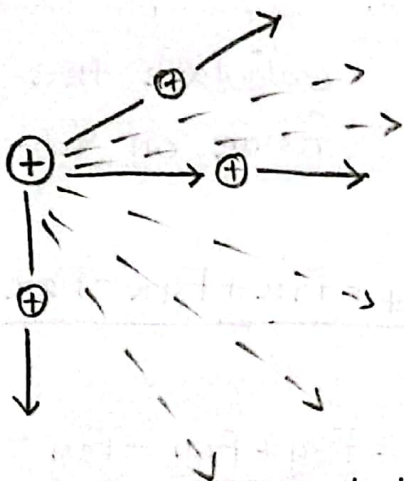


$$F_{x\text{Net}} = -F_{1x} - F_{2x} = F_1 \cos \theta - F_2$$

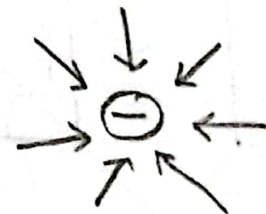
$$\begin{aligned} F_{y\text{Net}} &= -F_{1y} \\ &= -F_1 \sin \theta \end{aligned}$$

Physics, class: 16

21.11.2023.



Outward



Inward

Electric Field: An electric field is a vector field that can be associated with each point in space, the force per unit charge exerted on a positive test charge at rest at that point.

Electric Field Line: Electric field lines are the way of visualizing electric fields.

Electric Field

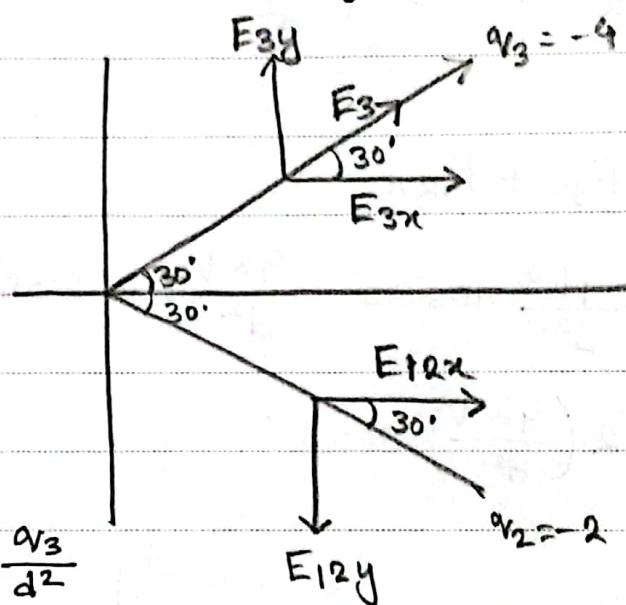
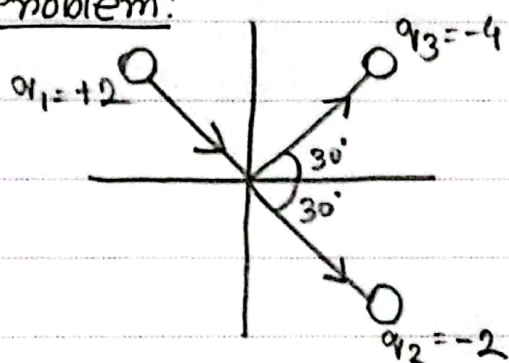
$$\vec{E} = \frac{\vec{F}}{q_0}$$

$$\vec{F}_1 + \vec{F}_2 = 0$$

$$\vec{E}_1 + \vec{E}_2 = 0$$

* Show that there is no place between two positive charges where a positive test charge will experience no force.

Problem:



$$E_1 = \frac{kq_1}{d^2}, E_2 = k \frac{q_2}{d^2}, E_3 = k \frac{q_3}{d^2}$$

দুটি বৈদ্যুতিক প্রকার দিগন্ত হল তাদের resultant হবে
তাদের যোগফল,

$$E_{12} = E_1 + E_2$$

$$= \frac{k}{d^2} (q_1 + q_2)$$

$$= \frac{k}{d^2} (2q + 2q)$$

$$= k \frac{4q}{d^2}$$

$$E_{3x} = E_3 \cos 30^\circ = \frac{4kQ}{d^2} \cos 30^\circ$$

$$E_{3y} = E_3 \sin 30^\circ = \boxed{\frac{4kQ}{d^2} \sin 30^\circ}$$

$$E_{12x} = E_{12} \cos 30^\circ = 4 \frac{kq}{d^2} \cos 30^\circ$$

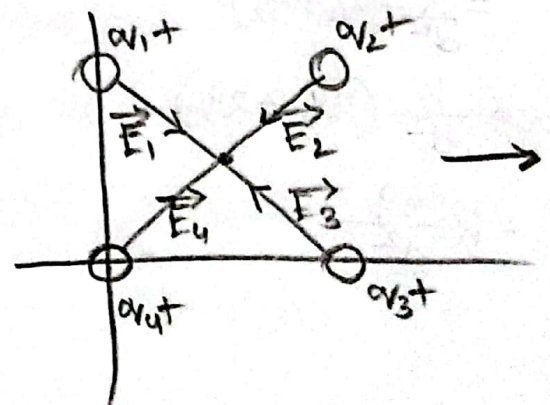
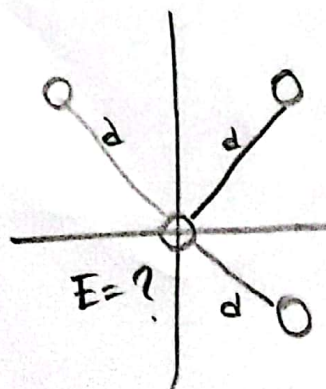
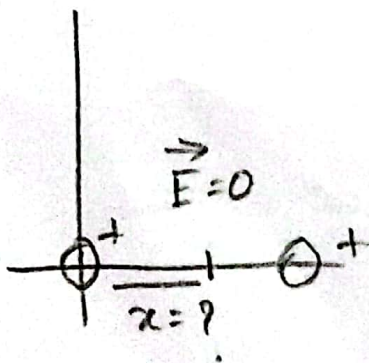
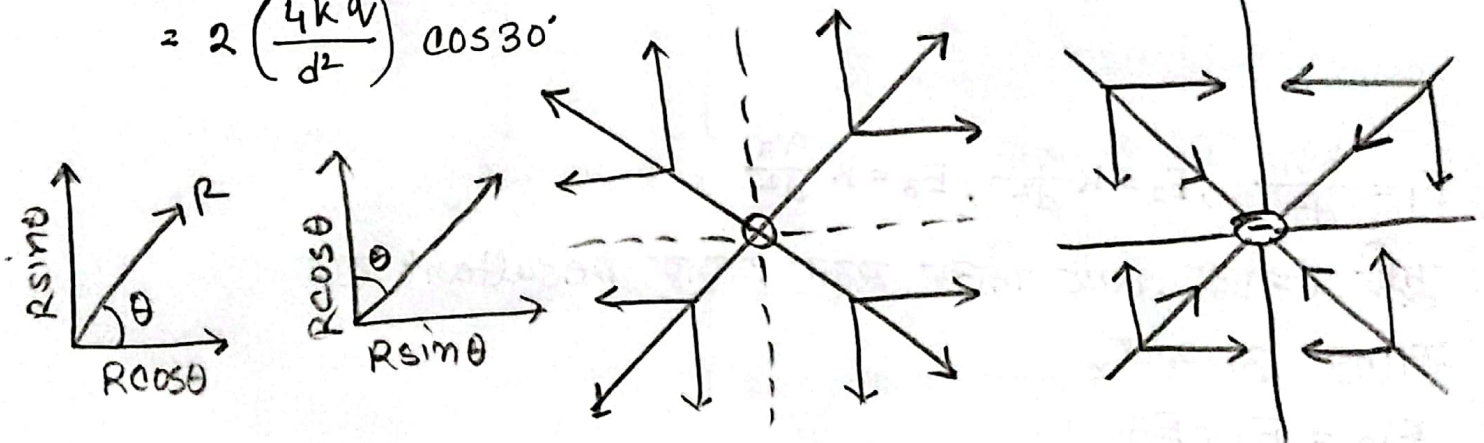
$$E_{12y} = E_{12} \sin 30^\circ = \boxed{4 \frac{kq}{d^2} \sin 30^\circ}$$

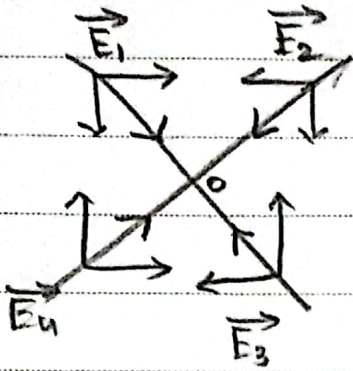
same value
different direction

$$E_{x\text{Net}} = E_{3x} + E_{12x}$$

$$= \frac{4kq}{d^2} \cos 30^\circ + \frac{4kq}{d^2} \cos 30^\circ$$

$$= 2 \left(\frac{4kq}{d^2} \right) \cos 30^\circ$$





$$E_{xNet} = E_{1x} - E_{2x} - E_{3x} + E_{4x}$$

$$E_{yNet} = -E_{1y} - E_{2y} + E_{3y} + E_{4y}$$

$$E_{Net} = \sqrt{E_{xNet}^2 + E_{yNet}^2}$$

$$E_{1x} = E_1 \cos \theta$$

$$\theta = \tan^{-1} \left(\frac{E_{yNet}}{E_{xNet}} \right)$$

$$E_{1y} = E_1 \sin \theta$$